

Explaining Variability in Shell-Based Oxygen Isotope Ratios in the Western North Atlantic Using a Proxy System Model

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The Western North Atlantic region has recently undergone rapid ocean warming, which has already affected key species. To investigate whether this warming trend is a recent phenomenon or something long-term, it is necessary to look back further than modern instrumentation allows. Paleoclimate proxies like *Arctica islandica* (ocean quahog) shells offer a look into past environmental conditions via shell-derived oxygen isotopic records. However, when looking at isotopic records from six different sites in the Western North Atlantic ranging from the Delmarva Peninsula north to the northern Gulf of Maine, large amounts of variability exist between these regions. To investigate these differences, climate model output and ocean reanalysis data were used to generate a simulated clam oxygen isotope record, which was then compared to the measured oxygen isotopes records between the 1850s and present day. This approach, called proxy system modeling, allows certain environmental variables to be held constant so that the factors that might be driving the observed oxygen isotopic differences between the sites can be isolated. Using both gridded reanalysis environmental data and ocean model climate outputs, the relationships were investigated between the pseudodata and shell records. Differences between the climate products as well as the effect of growing season, temperature, and seawater isotope content were also explored. For the reanalysis products, both SODA (0.5° resolution) and GLORYS data (1/12° resolution) was used, while the ocean models used were iCESM (1.9°x2.5° resolution) and VIKING20X (1/20° resolution Atlantic Ocean nested in a 1/4 ° global ocean). The iCESM model is the isotope-enabled version of the fully coupled, global Community Earth System Model (CESM1.2), while VIKING20X uses the NEMO3.6 configuration and is forced by the JRA-55 atmospheric product from 1958-present. Using spectral analysis and measures of coherence, periodicity was examined, as were the differences in how the periodicity behaves across different model inputs and sites. In doing so, a better understanding of the specific environmental conditions that influence the oxygen isotope records from each site is gained, enabling a more robust reconstruction of past environmental changes along the western North Atlantic shelf.